Continuum Damage Mechanics

Course: Composite Material Damage Mechanics

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General Concept

1 Introduction

- **Limitation of classical continuum mechanics**: Microdefects (grains, dislocations) are heterogeneous, but damaging processes involve strain localization and irreversibility.
- **Origin**: Kachanov (1958) and Rabotnov (1969) on metal creep $\rightarrow$ macroscopic damage variables.
- **Framework**: Thermodynamics of irreversible processes; complements Fracture Mechanics.
- **Goal**: Review some general features of C.D.M. and to summarize its main possibilities and introduce the possibility of describing the coupling effects between damage processes and the stress-strain behavior of materials.

- **Key aspects**:
  1. Definitions and measures of damage (mechanical variables for microstructure).
  2. Mechanical behavior of damaged material (includes anisotropy).
  3. Evolution equations for damage variables.

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2 Damage Measures and Definitions

2.1. The Problem of Crack Initiation

- **Damage Variable:** Not directly measurable; requires indirect procedures & corresponding models.
- **Crack Initiation (Jeal, 1985):** "Breaking up" of the continuum volume element.
  - Arbitrary in fatigue due to complex microstructural interactions.
- **Fracture Mechanics Requirement:** Macroscopic crack must span several grains for homogeneity (size, geometry, direction; Fig. 1).

2.1. The Problem of Crack Initiation

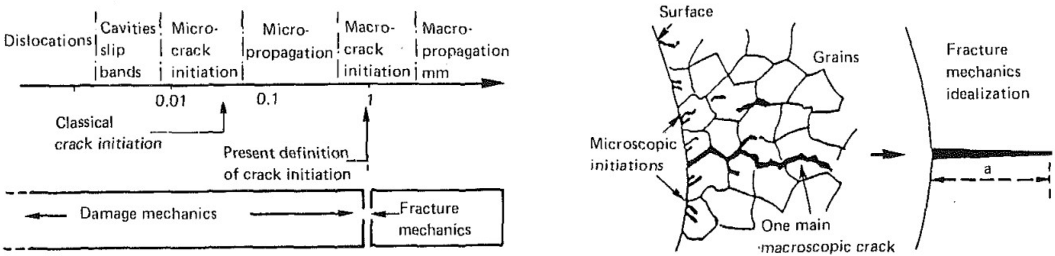


Fig. 1: (a) Schematics of the fatigue crack growth; (b) illustration of a macroscopic crack initiation concept.

2.2. The Remaining Life

- **Engineering Definition (Manson, 1979; Chaboche, 1974):** Damage measured via remaining life.
- **Miner's Linear Rule:**

$$\frac{N_2}{N_{F_2}} = 1 - D = 1 - \frac{N_1}{N_{F_1}} \tag{2}$$

- **Key Observation (Krempl, 1977):** Damage evolution is not unique; depends on loading level (unseparability; Fig. 2).
- **Limitation:** Remaining life does not uniquely determine D (one-to-one mapping ambiguity).
- **Applicability:** Valid for both fatigue and creep.

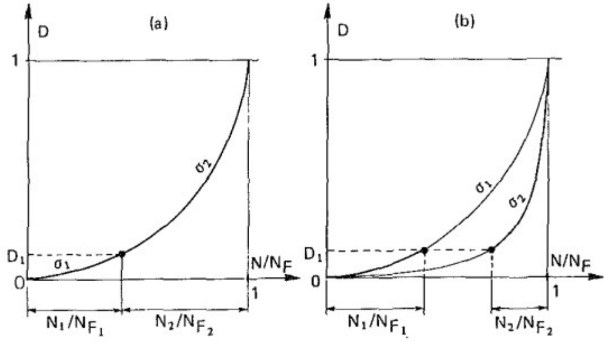


Fig. 2: Schematic of damage evolution curves as deduced from remaining life measures

- **Definition Basis (Dyson and McLean, 1977; Levaillant and Pineau, 1982):** Direct observation of defects and quantify the irreversible defects.
- **Limitations:**
  - Destructive measuremen,
  - early-stage damage hard to observe,
  - Requires mapping to macroscopic variables for structural analysis.
- **Net Area Reduction (Scalar):**

$$D_n = \frac{S_D}{S} = \frac{S - S^*}{S} \quad (0 \leq D_n \leq D_c) \tag{3}$$

  - S : total area, S^* : effective area
  - **Critical Value:** $0.2 < D_c < 0.8$ for metals (Fig. 3).

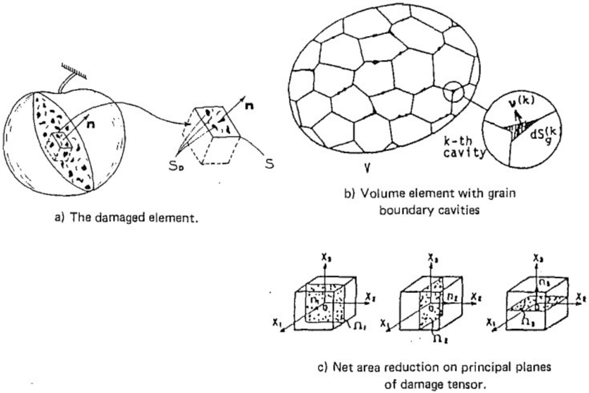


Fig. 3: Net area reduction: (a) The damaged element; (b) volume ele- ment with grain boundary cavities; (c) net area reduction on principal planes of damage tensor

- **Net Stress Concept (Murakami & Ohno, 1980):** For creep damage in polycrystalline metals, cavities on grain boundaries reduce the effective load-bearing area.
  - Damage state described by a **second-rank symmetric tensor $\boldsymbol{\Omega}$** (Murakami, 1983):

$$\boldsymbol{\Omega} = \frac{3}{S_g(V)} \sum_{k=1}^{N} \int_V \vec{v}^{(k)} \otimes \vec{v}^{(k)} \, dS_g^{(k)} \tag{4}$$

    where $dS_g^{(k)}$ = area of boundary occupied by k-th cavity, $\vec{v}^{(k)}$ = unit normal vector.
  - The net stress σ^* is defined by assuming equivalence between damaged and undamaged material (Fig. 4).
- **Anisotropic Damage:**
  - The tensor definition naturally accounts for directional effects.
  - Alternative approach (Krajcinovic, 1983): family of vectors, each associated with a typical microcrack direction (similar to slip theory of plasticity).

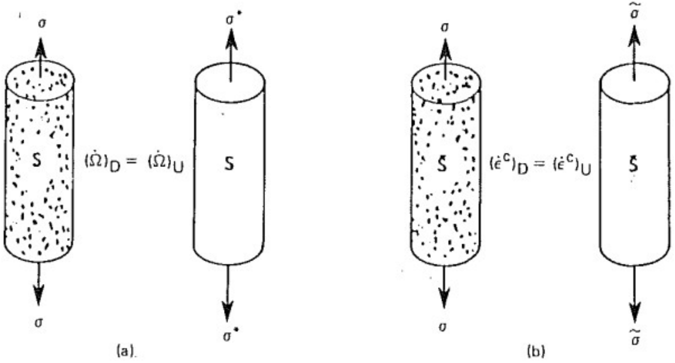


Fig. 4: (a) Net stress tensor for damage growth; (b) effective stress tensor

2.4. Physical Parameters and the Effective Stress Concept

- **Physical quantities affected by damage:**
  - Density change (Jonas & Baudelet, 1977) – ductile failure,
  - Resistivity change (Cailletaud et al., 1980) – similar to mechanical measures,
  - Acoustic emission, sound velocity,
  - fatigue limit change (Bui-Quoc et al., 1971) – related to remaining life.
- **Effective stress concept (Chaboche, 1974):**

$$\bar{\sigma} = \sigma \frac{S}{\bar{S}} = \frac{\sigma}{1 - D} \tag{5}$$

  - with homogenization support (Duvaut, 1976).

2.4. Physical Parameters and the Effective Stress Concept

- **Ductile rupture (Lemaitre, 1985):** Measure elastic modulus change.

$$\sigma = \tilde{E}\,\epsilon_e, \quad \bar{\sigma} = E\,\epsilon_e \;\Rightarrow\; D = 1 - \frac{\tilde{E}}{E} \tag{6}$$

- **Brittle creep damage (Lemaitre & Chaboche, 1978):** Secondary creep power law (Fig. 5(a)):

$$\dot{\epsilon}_s = \left(\frac{\sigma}{\lambda}\right)^N \;\Rightarrow\; D = 1 - \left(\frac{\dot{\varepsilon}}{\dot{\varepsilon}_s}\right)^{1/N} \tag{7,8}$$

2.4. Physical Parameters and the Effective Stress Concept

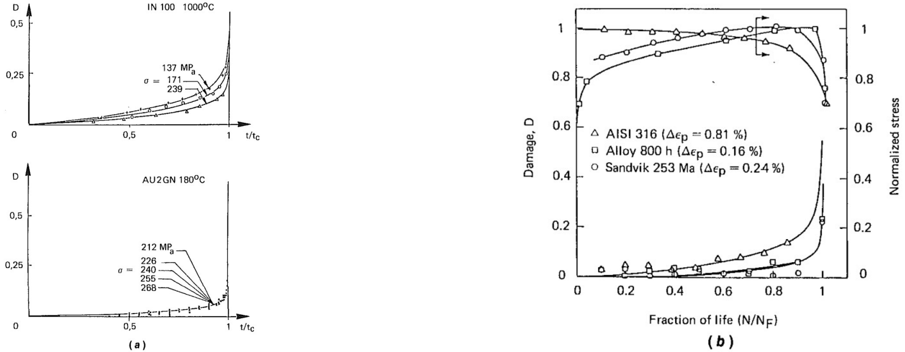
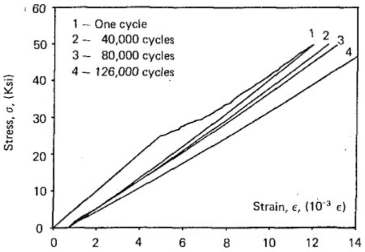


Fig. 5: (a) Creep damage measures—alloys IN 100 at 1000°C and AU2GN at 180°C; (b) damage accumulation curves as measured from the change in elastic response with corresponding cyclic stress response.

2.4. Physical Parameters and the Effective Stress Concept

- **Fatigue damage:** Measures are more complex due to:
  - Surface localization, cyclic hardening/softening (not direct damage).
  - Effective stress concept correlates elastic modulus drop with peak stress evolution (Plumtree & Nilsson, 1986).
  - Good correlation with quantitative microcrack evaluation (Cailletaud & Levaillant, 1984; Hua & Socie, 1984; Fig. 5(b)).
- **Damage in composites (graphite/epoxy laminates, Charewicz & Daniel, 1985):**
  - Mechanisms: fiber debonding, matrix microcracking, delamination.
  - Elastic modulus drops on first cycle (partial debonding), then progressively decreases during fatigue with stabilization (Fig. 6).
  - Final delamination leads to rupture.
  - Clear correspondence: modulus decrease, residual strength drop, and remaining life.

2.4. Physical Parameters and the Effective Stress Concept



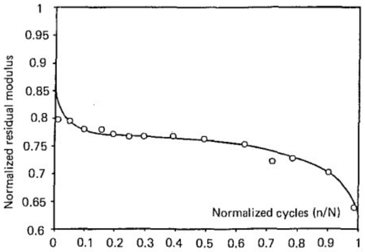


Fig. 6: Fatigue of $[0/904]_s$ graphite/epoxy specimens, σ_{max} = 345 MPa (50 ksi), R = 0.1, from Charewicz and Daniel (1985). (a) stress-strain curves at various stages of the fatigue life; (b) normalized residual elastic modulus.

2.4. Physical Parameters and the Effective Stress Concept

- **Active and Passive Damage:**
  - Damage can be either active or passive.
  - When defects, especially microcracks, are closed, they do not affect macroscopic behavior; such damage is considered passive (e.g., under compression).
- **Probabilistic Nature of Damage:**
  - Damage exhibits a probabilistic nature at the microscale.
  - When considering the residual strength of the material, this leads to statistical definitions of damage variables.

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3 Thermodynamic Aspects

3. Thermodynamic Aspects
Internal State Variables

- **Thermodynamic Aspects:** Based on thermodynamic theory of irreversible processes with internal state variables (Sidoroff, 1975; Germain et al., 1983).
- **Simplified case:** small strain, isotropic hardening, isotropic damage.
- **Extensions:** kinematic hardening (Halphen and Nguyen, 1975); anisotropic damage (Chaboche, 1978; Lemaitre and Chaboche, 1985); finite strain (Rousselier, 1980).

Table 1: State Variables

Observable variable	Internal variables	Associated variables
Elastic strain tensor ϵ_e		Stress tensor σ
Temperature T		Entropy S
	Isotropic hardening r	Size increase of yield surface R
	Damage D	Damage strain energy release rate Y

3. Thermodynamic Aspects

Plastic Strain Definitions

- **Plastic strain tensor from total strain:**

$$\epsilon_p = \epsilon - \epsilon_e$$

- **Accumulated plastic strain:**

$$p = \left(\frac{2}{3}\,\dot{\epsilon}_p : \dot{\epsilon}_p\right)^{1/2}$$

3.1. Thermodynamic Potential

Uncoupled Elasticity and Plasticity

- **Specific free energy Ψ as thermodynamic potential:**

$$\Psi = \Psi_e(\epsilon_e, T, D) + \Psi_p(T, r) \tag{9}$$

- **Damaged elastic behavior:**

$$\Psi_e = \frac{1}{2\rho}(1 - D)\Lambda : \epsilon_e : \epsilon_e \tag{10}$$

- **Stress:**

$$\sigma = \rho\frac{\partial\Psi}{\partial\epsilon_e} = (1 - D)\Lambda : \epsilon_e \quad \text{or} \quad \sigma = \tilde{\Lambda} : \epsilon_e \tag{11}$$

3.1. Thermodynamic Forces

- **Thermodynamic forces:** Associated with D and r:

$$Y = \rho \frac{\partial \Psi}{\partial D} = -\frac{1}{2} \Lambda : \epsilon_e : \epsilon_e, \qquad R = \rho \frac{\partial \Psi}{\partial r} \tag{12}$$

  — D includes all damaging effects (constant density). In finite strains (Rousselier, 1980), density change can be used as damage variable.
  — Complete separability of hardening and damage: Ψ_p independent of D.

- **Elastic strain energy release rate (Chaboche, 1977):**

$$-Y = \frac{W_e}{1-D} = \frac{1}{2} \frac{dW_e}{dD} \quad \text{at constant } \sigma \text{ and } T \tag{13}$$

  — Expression in terms of stress invariants (Lemaitre, 1984):

$$-Y = \frac{\sigma_{eq}^2}{2E(1-D)^2} \left[\frac{2}{3}(1+\nu) + 3(1-2\nu) \left(\frac{\sigma_H}{\sigma_{eq}} \right)^2 \right] \tag{14}$$

- **Total dissipated energy separates into:**
  1. Energy stored in the system (hardening).
  2. Heat dissipated energy.
  3. Energy released during damaging $-Y\,\delta D$, eventually converted into heat.

  — The stored energy equals work above initial yield limit – consequence of simplifying assumptions on isotropic hardening.

  — Additional energies converted to heat can be described using nonlinear kinematic hardening (Lemaitre and Chaboche, 1985) and specific choices for Ψ_p(Fig. 7).

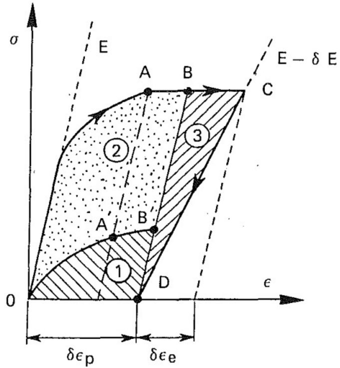

Fig. 7: Schematic of the dissipation during plastic flow and damage growth

- **Second principle:** Intrinsic dissipation must be positive:

$$\sigma : \dot{\epsilon}_p - R\,\dot{r} - Y\,\dot{D} > 0 \tag{15}$$

  - Hardening and damage uncoupled $\Rightarrow$ sufficient condition:

$$-Y\dot{D} > 0 \tag{16}$$

  - Since $-Y$ is quadratic, this implies $\dot{D} > 0$.
  - Rupture criterion: $-Y = |Y| = Y_c$ (crack initiation) – elastic energy criterion.

- **Rupture stress σ_R in 1D:**

$$Y_c = \frac{\sigma_R^2}{2E(1-D_c)^2} \;\Rightarrow\; D_c = 1 - \frac{\sigma_R}{(2EY_c)^{1/2}} \tag{17}$$

  - Experimental range: $0.2 < D_c < 0.8$ – allows neglecting $(1-D_c)^x$ for $x \gg 1$.

3.2. Dissipation

- **Dissipation potential ϕ:** Scalar convex function of fluxes $(\dot{\epsilon}_p, \dot{r}, \dot{D}, q)$ or dual variables via Legendre-Fenchel transform (Germain et al., 1983):

$$\phi(\sigma, R, Y; \epsilon_e, T, p, D)$$

- **Constitutive equations for dissipative variables:**

$$\dot{\epsilon}_p = \frac{\partial \phi^*}{\partial \sigma}, \qquad \dot{r} = -\frac{\partial \phi^*}{\partial R}, \qquad \dot{D} = \frac{\partial \phi^*}{\partial Y} \tag{18}$$

  — If ϕ^* convex in $-Y$, damage dissipation (16) automatically positive (Chaboche, 1977).

3.2. Dissipation

- **Time-independent plasticity, isotropic hardening (Lemaître and Chaboche, 1985):**

$$f(\sigma, R, D) = \bar{\sigma}_{eq} - R - k < 0, \qquad \bar{\sigma}_{eq} = \frac{\sigma_{eq}}{1-D} \tag{19}$$

- **Normality rule (σ': stress deviator):**

$$\dot{\epsilon}_p = \lambda \frac{\partial f}{\partial \sigma} = \frac{3}{2} \frac{\lambda}{1-D} \frac{\sigma'}{\sigma_{eq}} \tag{20}$$

$$\dot{r} = \lambda \frac{\partial f}{\partial R} = \lambda = \left(\frac{2}{3} \dot{\epsilon}_p : \dot{\epsilon}_p \right)^{1/2} = (1-D)\, \dot{p}$$

  — Plastic multiplier from consistency $\dot{f} = 0$. For $D = 0$, $p =$ accumulated plastic strain.

  — Normality is sufficient, not necessary (Onat, 1985). Sometimes too restrictive (e.g., creep). Only required: $\dot{D} > 0$ (since $-Y > 0$).

- **Generalization to large strains with modifications (Rousselier, 1980):**
  - First damage parameter: density change – affects elastic behavior.
  - Second parameter: explicit dependency of plastic potential on hydrostatic stress – generalizes Rice & Tracey (1969) formula for spherical void growth under high triaxiality.

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Damage Growth, Crack Initiation, and Crack Growth

- **Damage variables** specific to different processes:
  - Creep, fatigue (high/low cycle), ductile damage, brittle damage.
- **Damage growth equations** derived from a dissipative potential (thermodynamics).
- **Two main application domains**:
  1. **Life prediction**: Damage growth and crack initiation in structures (creep, fatigue, creep-fatigue).
  2. **Macroscopic crack growth**: Improve "local approaches to fracture" beyond classical Fracture Mechanics.
- **Practical outcome**: Better prediction of material deterioration and residual life.

4.1. Creep Damage

- **Creep damage evolution:** Under pure tensile stress,

$$dD = \left(\frac{\sigma}{A}\right)^{r} (1 - D)^{-k}\, dt \tag{21}$$

  - Damage variable D: 0 (undamaged) to 1 (rupture).
  - Integration under constant stress gives rupture time:

$$t_c = \frac{1}{k+1} \left(\frac{\sigma}{A}\right)^{-r} \tag{22}$$

  - Damage evolution with time:

$$D = 1 - \left(\frac{t}{t_c}\right)^{1/(k+1)} \tag{23}$$

4.1. Creep Damage

- **Effective stress concept** (Chaboche): Norton's secondary creep law extended:

$$\dot{\epsilon}_p = \left[\frac{\sigma}{K(1-D)}\right]^n \tag{24}$$

- Damage depends on stress invariants:
  - Octahedral shear stress $J_2(\sigma)$ (shear effects)
  - Hydrostatic stress $J_1(\sigma)$ (cavity growth)
  - Maximum principal stress $J_0(\sigma) = \sigma_1$ (microcrack opening)(Fig. 8)
- **Hayhurst equivalent stress** (linear form):

$$\chi(\sigma) = \alpha J_0(\sigma) + \beta J_1(\sigma) + (1-\alpha-\beta)J_2(\sigma) \tag{25}$$

- Multiaxial time to failure:

$$t_c = \frac{1}{k+1}\left(\frac{\chi(\sigma)}{A}\right)^{-r} \tag{26}$$

- Alternative product form (physical interpretation):

$$\chi(\sigma) = [J_0(\sigma)]^{\alpha}[J_2(\sigma)]^{1-\alpha} \tag{27}$$

- Alternative damage growth equation using creep strain (Contesti, 1986):

$$dD = A\,[J_0(\sigma)]^{\alpha}\,\epsilon_{eq}^{\beta}\,d\epsilon_{eq} \tag{28}$$

- One-to-one mapping between strain-based and time-based formulations (Eqs. (21) and (24)):

$$D \;\longrightarrow\; \left[1-(1-D)^{k-n+1}\right]^{\beta+1} \tag{29}$$

- **Note:** Pure compression yields no damage under product form – not fully accurate for polycrystalline metals (two-level creep tests, Policella et al., 1982).

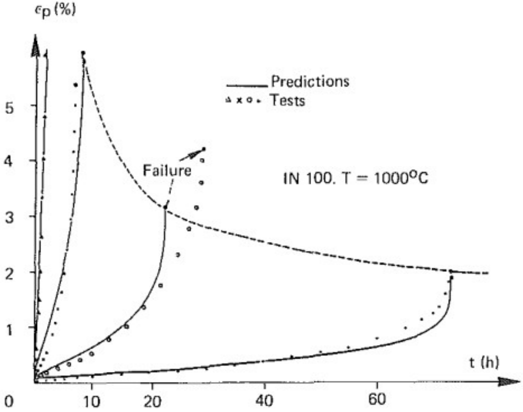

Fig. 8: Calculated and measured creep curves on superalloy IN 100. Prediction of creep ductility.

4.2. Fatigue Damage

- Key aspects of fatigue damage:
  - Microinitiation and micropropagation stages.
  - Nonlinear cumulative effects (two-level tests, block-program loading).
  - Decrease of fatigue limit after prior damage.
  - Mean-stress effects on S–N curves and fatigue limit.
- General damage accumulation model (Chaboche, 1974):

$$dD = D^{\alpha(\sigma_M, \bar{\sigma})} \left[\frac{\sigma_M - \bar{\sigma}}{M(\bar{\sigma})} \right]^{\beta} dN \tag{30}$$

  where σ_M = maximum stress, $\bar{\sigma}$ = mean stress.
- Two-level test integration gives nonlinear interaction:

$$\frac{N_2}{N_{F_2}} = 1 - \left(\frac{N_1}{N_{F_1}} \right)^{\frac{1-\alpha_2}{1-\alpha_1}} \tag{31}$$

- Function $M(\bar{\sigma})$ derived from linear dependency between $\bar{\sigma}$ and fatigue limit.

4.2. Fatigue Damage

- **Equivalence of damage definitions:**
  - Effective stress concept (CDM).
  - Remaining life concept (Eq. (30) based).
  - Physical microcrack measurements (Cailletaud & Levaillant, 1984; Socie et al., 1983).
- **Low-Cycle Fatigue:** Life parametrized by plastic (or total) strain range.

  One-to-one relation between σ_M and $\Delta\varepsilon_p$ (cyclic curve).

  Mean-stress influence naturally included in Eq. (30).
- **Multiaxial generalization:**
  - Two parameters needed: equivalent shear-stress amplitude, and mean/maximum hydrostatic stress.
  - Near fatigue limit: independence of mean shear stress and hydrostatic range (Dang Van, 1973).
  - Nonproportional loading conditions require further study.

4.3. Creep-Fatigue Interaction

- **Approach (Chaboche, 1980):** Direct summation of creep and fatigue damage rates.

$$dD = f_c(\sigma, D)\,dt + f_F(\sigma_M, \bar{\sigma}, D)\,dN \tag{32}$$

- **Functions:**
  - f_c: from creep model (e.g., Kachanov-Rabotnov).
  - f_F: from fatigue model (e.g., nonlinear cumulative).
- **Prediction:** Integrate numerically for low-frequency or hold-time conditions.
- **Critical Clarification:**
  - Physical defects (cavities, microcracks) are **not** added.
  - Only **mechanical effects** added via effective stress concept.

4.4. Oxidatipn-Fatigue-Creep Interactions

- **High-Temperature Effect:** Oxidation superimposes on creep and fatigue.
- **Oxidation Enhances:**
  - **Nucleation:** Surface cracks, grain boundary oxidation, internal voids.
  - **Growth:** Crack tip deterioration, void growth from particles.
- **Modeling Integration:**
  - Explicit in fatigue models (initiation + propagation phases).
  - Implicit (conceptual) in creep damage equations.

4.5. Ductile Plastic Damage

- **Ductile damage:** initiation and growth of cavities due to large deformations.
- Dissipative potential in time-independent plasticity (Lemaitre & Chaboche, 1985):

$$\phi^* = \bar{\sigma}_{eq} - R - k + \frac{S}{2}\frac{1}{1-D}\left(-\frac{Y}{S}\right)^2 \tag{33}$$

- **Constitutive equations:**

$$\dot{\epsilon}_p = \lambda\frac{\partial\phi}{\partial\sigma}, \qquad \dot{r} = -\lambda\frac{\partial\phi^*}{\partial R} = \dot{p}(1-D) \tag{34}$$

$$\dot{D} = -\lambda\frac{\partial\phi^*}{\partial Y} = -\frac{Y}{S}\dot{p} \tag{35}$$

- **Hardening law:** Using the relation (14),

$$\sigma_{eq} = R + k + K\dot{p}^{1/M} \tag{36}$$

4.5. Ductile Plastic Damage

- Combining (14)–(16) gives the differential equation (Lemaitre, 1985):

$$\dot{D} = \frac{K^2}{2ES}\left[\frac{2}{3}(1+\nu) + 3(1-2\nu)\left(\frac{\sigma_H}{\sigma_{eq}}\right)^2\right] p^{2/M}\dot{p} \tag{37}$$

- For radial loading (constant triaxiality ratio σ_H/σ_{eq}):
  - Damage threshold: $p < p_D \;\Rightarrow\; D = 0$
  - Rupture strain: $p = p_R \;\Rightarrow\; D = D_c$
- Integrated form (neglecting elastic strain):

$$D = \frac{D_c}{\epsilon_R - \epsilon_D}\left\{p\left[\frac{2}{3}(1+\nu) + 3(1-2\nu)\left(\frac{\sigma_H}{\sigma_{eq}}\right)^2\right] - \epsilon_D\right\} \tag{38}$$

- Identification from experiments: ϵ_D, ϵ_R, D_c (Poisson's ratio ν known).
- Model predicts influence of triaxiality on strain to failure (Fig. 9).
- Applicable to fracture limits in metal forming (Lemaitre, 1984).

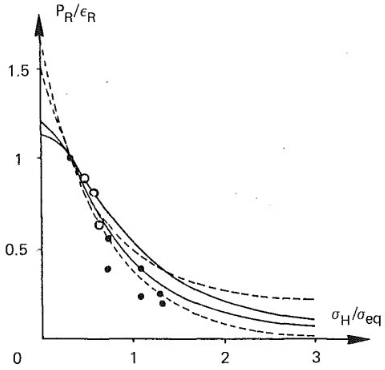

Fig. 9: Influence of triaxiality on fracture strain. •• : A508 steel; ∘∘ : H steel; ===== : domain covered by McClintock/Rice-Tracey models; ===== : domain covered by the present model.

- Simple damage growth law for pure tension (Lemaitre & Chaboche, 1985):

$$dD = \begin{cases} \left(\dfrac{\epsilon}{\epsilon_0}\right)^s d\epsilon & \text{if } \epsilon = \xi,\ d\epsilon = d\xi > 0 \\ 0 & \text{if } \epsilon < \xi \text{ or } d\epsilon < 0 \end{cases} \tag{39}$$

- Integration with elastic behavior ($\sigma = E(1 - D)\epsilon$) gives:

$$D = \left(\frac{\epsilon}{\epsilon_R}\right)^{s+1}, \qquad \sigma = E\epsilon\left[1 - \left(\frac{\epsilon}{\epsilon_R}\right)^{s+1}\right] \tag{40}$$

where ϵ_R = rupture strain ($D = 1$).

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5 Life Prediction in Structures

- **Two independent aspects for life calculations at high temperature:**
  - Macrocrack initiation – essential for design methodologies.
  - Crack propagation – "Damage Tolerance" concept for design and maintenance.
- **Classical approach:**
  - Compute stress-strain state at each point.
  - Predict crack initiation from stress/strain fields.
- **Problem:** fatigue damage nucleation/growth from initial stress concentrations. Macroscopic gradients vs. microlevel.

- **Overcoming strategies:**
  - Macro- and micro-plasticity inducing stress redistribution.
  - Introduce a critical distance (material dependent).
- **Future approaches:**
  - Coupling between material deterioration and stress-strain behavior (Marquis et al., 1981).
  - No stabilized cycle – model follows physical processes throughout the whole life (thousands of cycles).
  - Currently impractical for real structures under fatigue or HT-LCF with hold periods, but progress in computer speed and "cycle jump" algorithms (Savalle & Culié, 1978) may enable future applications.
- **Creep situation:** Coupling already applied (Marquis et al., 1981; Hayhurst, 1983), improving life predictions.

5.2 Crack Propagation by Fracture Mechanics Concepts

- **Classical approach:**
  - Predict crack growth and fracture in structural components is based on Fracture Mechanics concepts.
  - Gloabl parameters like The stress intensity factor K, the elastic strain energy release rate G, the contour integral J,
  - Linear Elastic Fracture Mechanics constitutes a practical tool to predict crack growth.
  -
- **Three developments for high temperature problems:**
  - Creep-Fatigue Interaction for Brittle Materials,
  - Application Under Thermal Fatigue,
  - Application to Ductile Materials (Fig. 10).

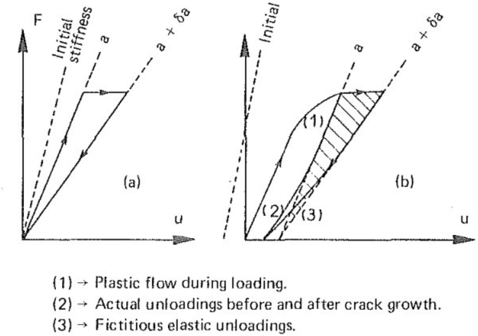

Fig. 10: Schematics of the elastic energy release rate: (a) elastic brittle solid; (b) elasto-plastic brittle solid.

5.3. Local Approaches to Fracture

- **Limitations of global Fracture Mechanics concepts:**
  - Complex situations: short cracks, history effects (overloads, warm-prestress), ductile fracture, creep crack growth.
  - Even when correlations exist, alternative methods are needed.
- **Local approaches (Janson & Hult, 1977):**
  - Consider actual behavior at crack tip.
  - Calculate local stress/strain fields and deterioration.
  - Local failure criterion predicts crack increment and instability.
- **Two levels of local approach:**
  1. Numerical methods with discrete crack increments (node release).
  2. Continuum Damage Mechanics (CDM) with progressive deterioration.

Level 1: Discrete Crack Increments

- **Technique:** Node release to produce crack growth when critical value of a physical quantity is reached at a critical distance ahead of the crack tip (Newman, 1974; Hinnerirhs, 1980; Anquez, 1983; Devaux & Mottet, 1984).
- **Physical quantity can be:**
  - Equivalent stress or strain.
  - Energy.
  - Local damage measure.
- **Mesh dependence:** Prediction depends on chosen mesh size (Newman, 1974). Local mesh size fixed via statistical considerations (Rousselier, 1986; Mudry, 1986).
- **Simplified approaches:** Use analytical stress-strain fields near crack tip (Maas & Pineau, 1985).

Level 2: Continuum Damage Mechanics

- **Concept:** Progressive deterioration with stiffness reduction. Crack = locus of material points with no rigidity (damage $\rightarrow$ critical value; Fig. 11).
- **Advantage:** No need for critical distance or node release. Stress redistribution due to viscoplasticity and strength decrease; stresses in plastic zone decrease continuously as damage increases approaching crack tip.
- **Pioneering work:** Hayhurst et al. (1975) in creep domain.
- **Current developments in:**
  - Ductile fracture (Rousselier et al., 1985; Rousselier, 1986).
  - Creep crack growth (Saanouni et al., 1985; Hayhurst, 1985).
  - Fatigue (Ben Allal et al., 1984).
  - Concrete rupture (Mazars, 1985).

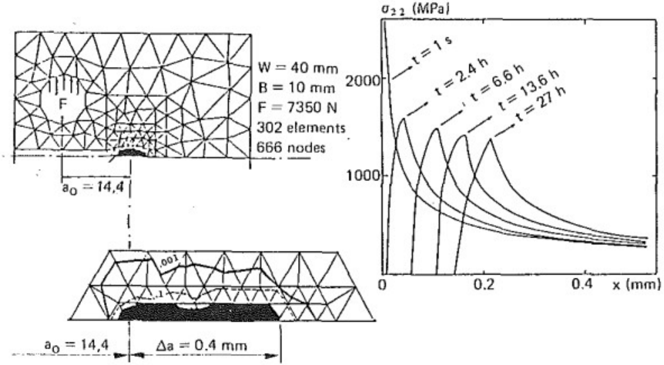

Fig. 11: Creep crack growth prediction by a local approach for a CT specimen in INCO 718 at 600 C.

5.3. Local Approaches to Fracture

- **Theoretical insight:** CDM with discontinuous damage growth law (sudden 0 to 1) solves Rice's paradox – consistency of energy dissipated at crack tip. Simplified damage laws allow analytical solutions (Bui & Ehrlacher, 1985; Bui et al., 1986).
- **Attractiveness:** Implement more physical rupture criteria and damage processes at crack tip.
- **Laboratory success:** Good results; beginning industrial application, especially for cleavage and ductile fracture (Devaux et al., 1986).

Remaining Problems

- **Computational cost:** However, difference not large if inelastic analysis already required for Fracture Mechanics. Simplified procedures (analytical fields) could offer intermediate practical way.
- **Mesh dependence:** Common feature of various local approaches. Crack width, growth rate, and failure load depend on chosen mesh size – even with full elastic-plastic-damage coupling (Saanouni et al., 1985).
- **Practical solution:** Local mesh size fixed per application after calibration.

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6 Conclusion

- **General concepts reviewed:** Measures and definitions of damage variables, incorporation into thermodynamic framework.
- **Main feature of CDM:** Coupling effects between damaging processes and stress-strain behavior.
- **Damage growth equations:** Specified and particularized to describe:
  - Creep and fatigue processes.
  - Ductile damage and brittle damage.

Conclusion

- **Practical damage rate equations presented** (uniaxial and multiaxial) for:
  - Creep, fatigue, creep-fatigue interaction.
  - Ductile plastic damage of metals.
  - Brittle damage (especially concrete).
- **Applications:** Predict crack initiation in structural components; CDM enables additional possibilities in local approaches to fracture.
- **Rational approach:** Extension of Continuum Mechanics – connects microstructure measurements to mechanical parameters.
- **Wide applicability:** Various loading conditions, materials, physical processes.
- **Life predictions:** Accounts for coupling between deterioration and mechanical behavior.
- **Future:** Will be developed further and extensively used as a complementary tool between Continuum Mechanics and Fracture Mechanics.

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7 references

References

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Q&A

Thank you for listening!

Your feedback will be highly appreciated!